

Bloom Energy: Another Renewable Energy Source

We have several renewable energy sources available such as solar, wind, hydro, bio-mass, thermal and tidal for generating electricity. But there is another energy called 'bloom energy' that can produce power

SYED ZAMEERUDDIN AND N. PAVAN KUMAR

The bloom energy server (commonly referred to as a Bloom Box) is a solid oxide fuel cell (SOFC) built by Bloom Energy, Sunnyvale, California, USA. It can use a wide variety of inputs (including liquid or gaseous hydrocarbons produced from bio-sources) to generate electricity on-site. It is highly efficient, low-cost and has lower polluting emissions.

The fuel cell can withstand temperatures up to 1800°F (such high temperatures can cause other types of fuel cells to break down), which is highly advantageous for its smooth operation. According to the company, a single cell (one 100mm×100mm metal alloy plate between two ceramic fast ion conductor plates) generates 25W.

To save money, the Bloom Energy server uses inexpensive metal alloy plates for electric conductance between the two fast iron conductor plates. In competing lower-temperature fuel cells, platinum is required at the cathode. Fuel cells can work in open atmosphere or in unassisted environment, but this leaves room for improvement. By operating in a pressurised environment, called pressurised hybrid system, efficiency can be improved.

The Bloom Box

The Bloom Energy server uses thin white ceramic plates (100mm×100mm) which are claimed to be made from beach sand. Each ceramic plate is coated with a green nickel oxide-based ink on one side (anode) and another black (probably

lanthanum strontium manganite) ink on the other side (cathode).

According to the *San Jose Mercury News*, "Bloom's secret technology apparently lies in the proprietary green ink that acts as the anode and the black ink that acts as the cathode," but in fact these materials are widely known in the field of SOFCs.

How an SOFC works

A fuel cell is like a battery that always runs. It comprises an electrolyte, an anode and a cathode. For an SOFC, the electrolyte is a solid ceramic material. The anode and cathode are made from special inks that coat the electrolyte. Unlike other types of fuel cells, no precious metals, corrosive acids or molten materials are required. Next, an electrochemical reaction converts fuel and air into electricity without combustion: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{electricity} + \text{heat}$

An SOFC is a high-temperature fuel cell. At high temperature, warmed air enters the cathode side of the fuel cell and steam mixes



Fig. 1: The Bloom Box



Fig. 2: Cathode and anode of the Bloom Box

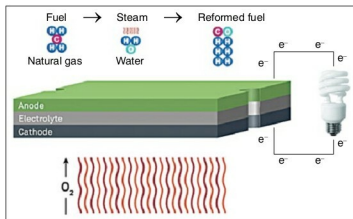


Fig. 3: Inside view of a solid oxide fuel cell (SOFC)